

universität freiburg



7TH ENZYMEML WORKSHOP · RÜDESHEIM · HANDS-ON SESSION 2 · 16 SEPTEMBER 2026

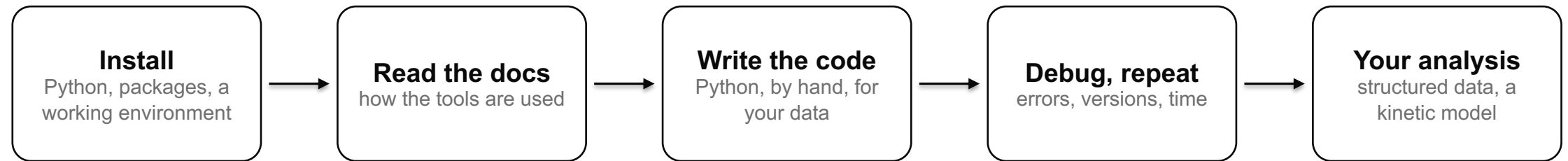
From measurement to kinetic modelling

Max Häußler

University of Freiburg

Last year the code was the hurdle

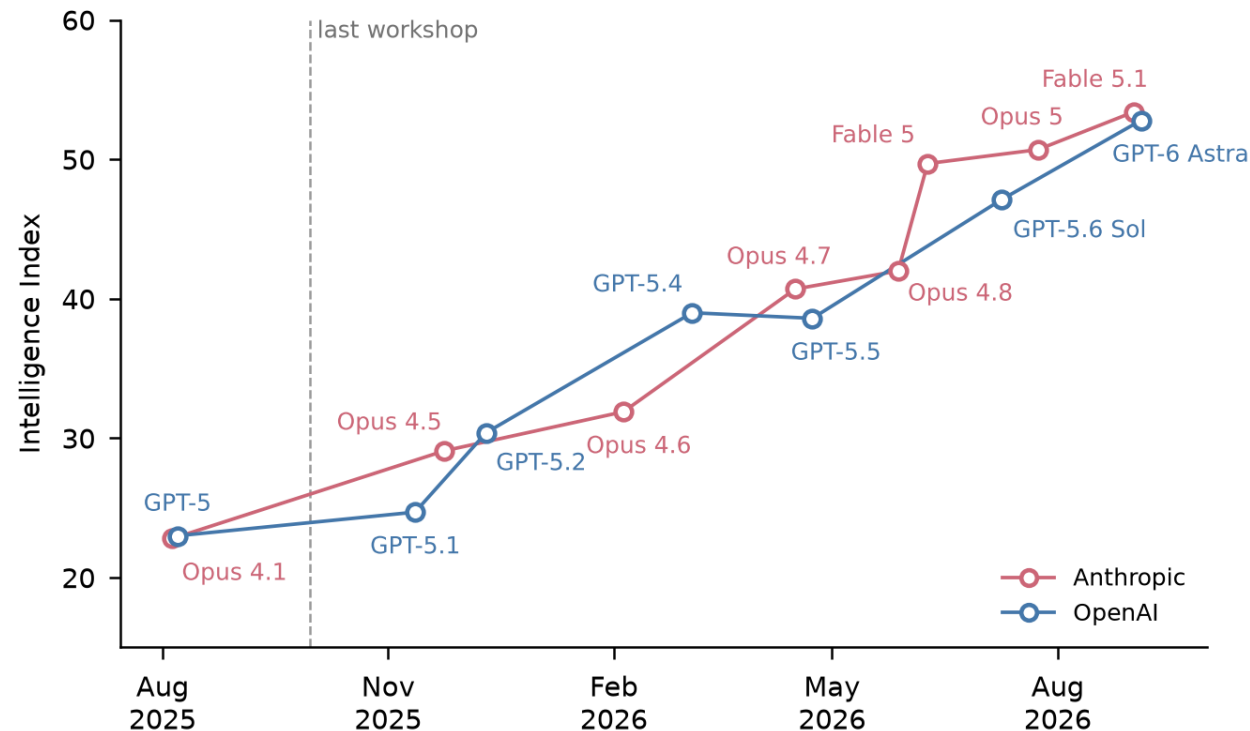
Previous hands-on sessions: we installed, read the docs and wrote the analysis code ourselves



- Failing installs and unfamiliar code: a hurdle for everyone who does not code daily
- So the data stayed unstructured, and the contemporary methods out of reach
- **Since then:** the models got better, and so did the harnesses that let them run tools

From 23 to 53 in one year

The two labs traded the lead several times; the biggest single step was Fable 5 in June 2026



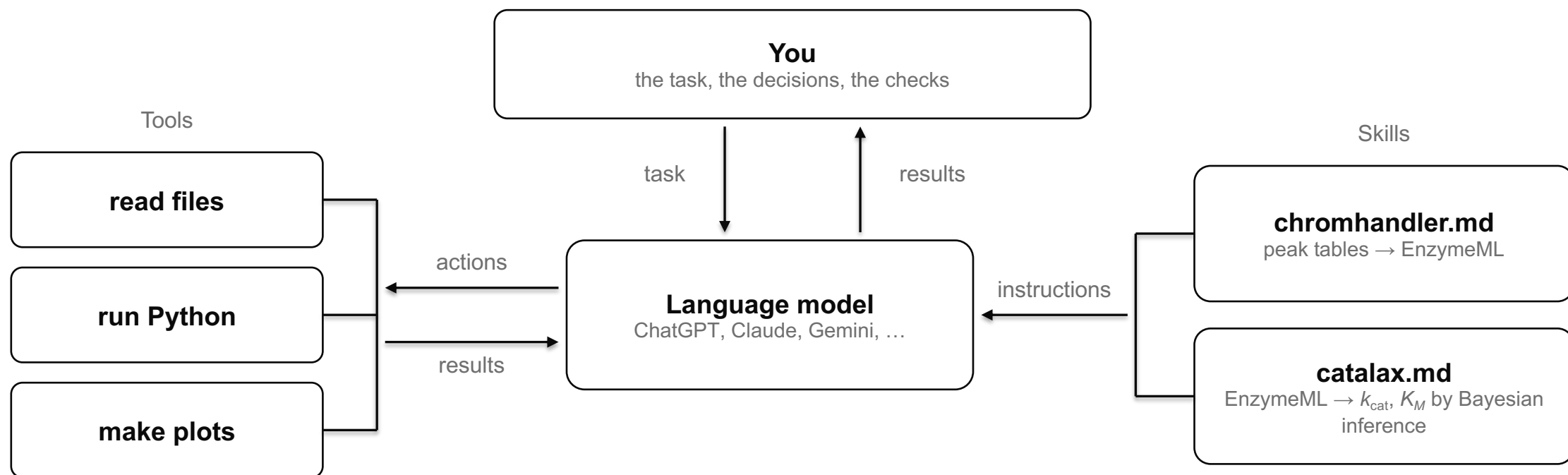
- The Intelligence Index is an independent score from 0 to 100 that combines many tests of knowledge, reasoning, coding and agentic work, with every model tested the same way.
- Shown are the flagship models of Anthropic and OpenAI that anyone can use, each at its strongest public setting.

Artificial Analysis Intelligence Index v4.3, artificialanalysis.ai; release dates from the vendors

The assistant types, you decide

- Today's models write correct code against our Python APIs from a one-page card
- They do not know where the peak is, which model is right, when a result is wrong
- So the scientist decides and checks; the assistant executes
- EnzymeML is the document that travels between the steps and the tools

Tools act, skills instruct, you decide

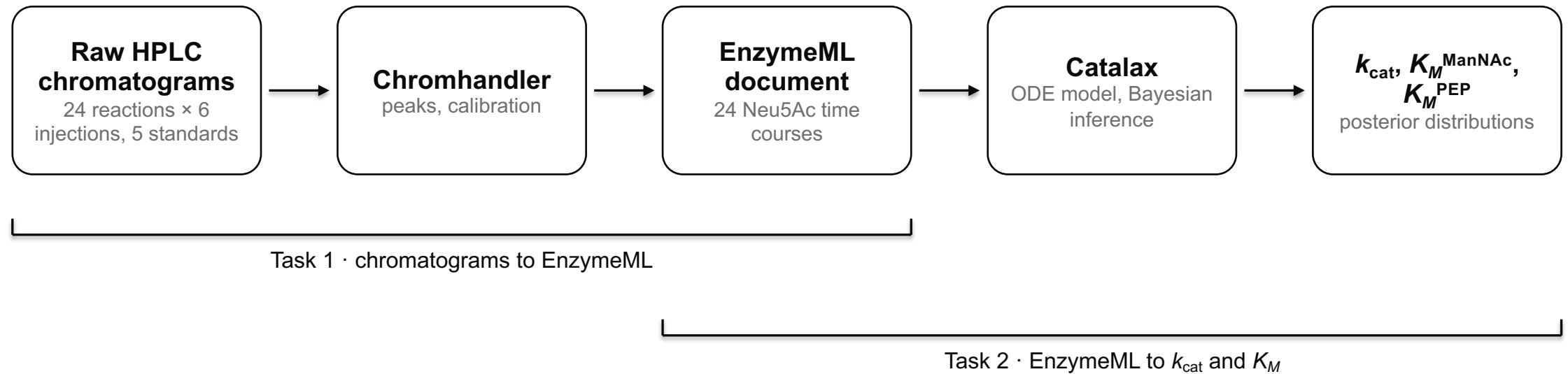


- Tools let the model read the data and run code; skills tell it how our libraries turn data into insight

WORKFLOW

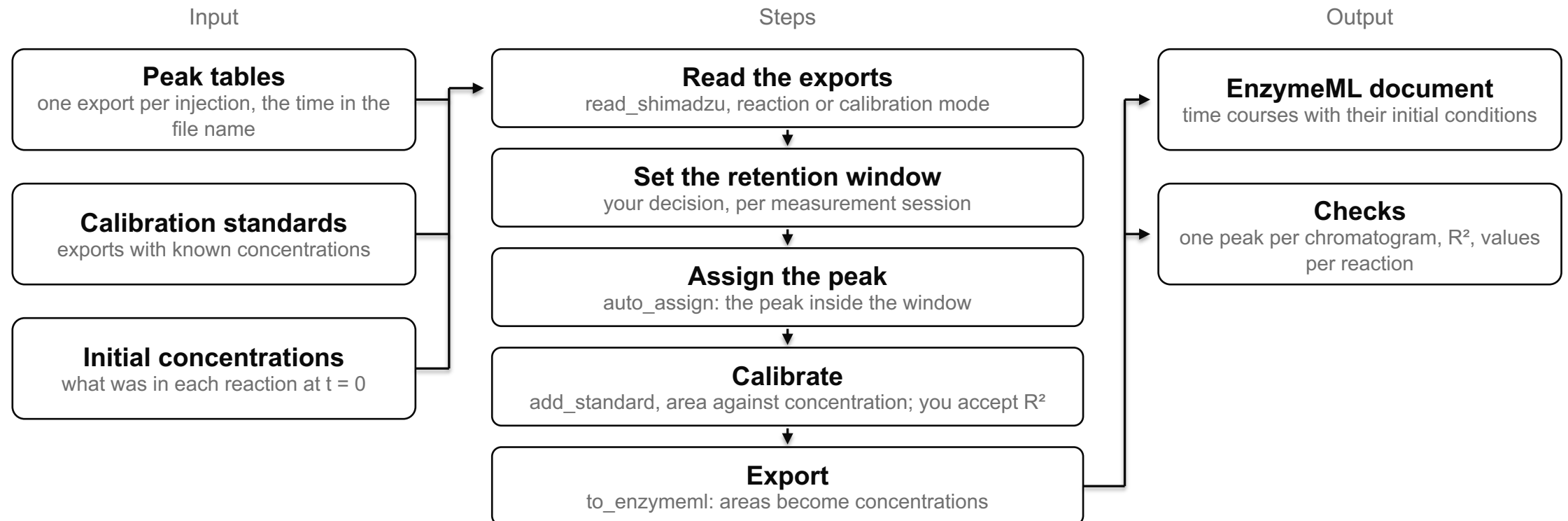
Two tasks, one document in between

Task 1 ends with the EnzymeML document; task 2 starts from it



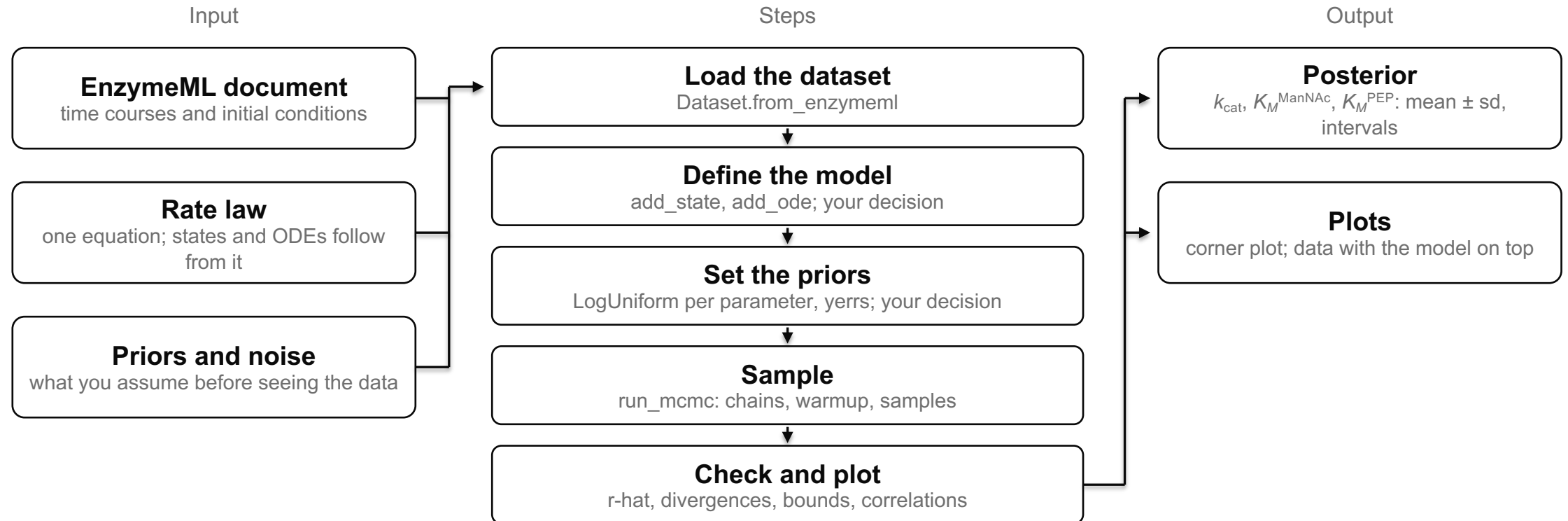
Chromhandler: from peak tables to concentrations

Reads the instrument's exports, assigns and calibrates one peak per compound, writes EnzymeML



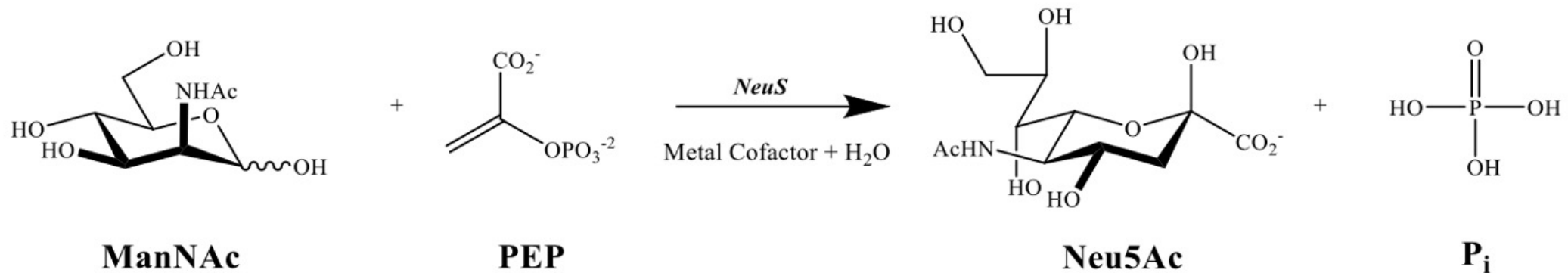
Catalax: from time courses to kinetic parameters

Builds the ODE model from a rate law and samples the posterior of its parameters



THE EXPERIMENT

ManNAc + PEP $\rightarrow$ Neu5Ac + Pi, followed by HPLC



- HPLC-PDA at 215 nm; only the product Neu5Ac is quantified
- ManNAc series: 13 reactions, ManNAc 0–100 mM at 20 mM PEP
- PEP series: 11 reactions, PEP 0–20 mM at 50 mM ManNAc
- Six injections per reaction; five calibration standards, 1–5 mM

Two series, one substrate varied at a time

	Samples	What varies	What is fixed
ManNAc series	13 reactions	ManNAc 0–100 mM	PEP 20 mM
PEP series	11 reactions	PEP 0–20 mM	ManNAc 50 mM
Calibration standards	5 standards	Neu5Ac 1–5 mM, with ManNAc and PEP	no enzyme

- 24 reactions sampled six times: 144 chromatograms
- Per injection you get the instrument's peak table, not the raw trace
- Initial concentrations are in conditions.csv

TODAY

What we do today

- **Task 1:** from the peak tables to an EnzymeML document (Chromhandler)
- **Task 2:** from the EnzymeML document to k_{cat} and K_M by Bayesian inference (Catalax)
- A live demo of each task first, then you work in pairs
- The checks in the task card say when you are done
- At the end we compare everybody's parameters and discuss

One repository, two ways to work

github.com/haeussma/enzymeml-workshop-modelling-2026 · start with the README

With a language model

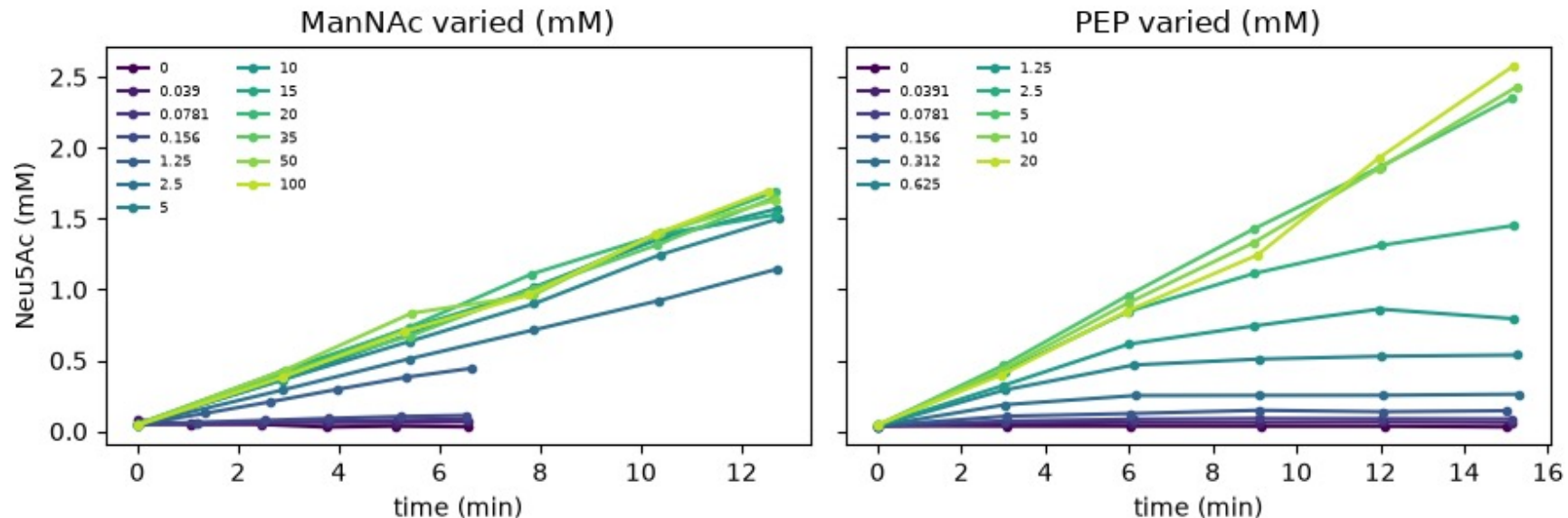
- **Any chat model**, free or by subscription: ChatGPT, Claude, Gemini, ...
- **Paste** the tool card and the task card; run the code, paste errors back
- **The checks** decide when you are done
- **notebooks/** are the reference solutions, checkpoints/ the result of task 1

Without writing code

- **Open the Colab link**: a prepared notebook, tools installed, data loaded
- **Run it cell by cell**; change a window, a prior, a series
- **Ask the model in Colab** to explain any cell

TASK 1

Task 1: from chromatograms to EnzymeML



- **Goal:** an EnzymeML document with 24 calibrated Neu5Ac time courses
- **Your decision:** the Neu5Ac retention window, different in standards and reactions
- **Checks:** slope $\approx 239\,000$ area per mM, $R^2 \geq 0.99$; 24 measurements, six increasing time points each
- **Live demo:** one deliberate mistake, the standards window used for the reactions

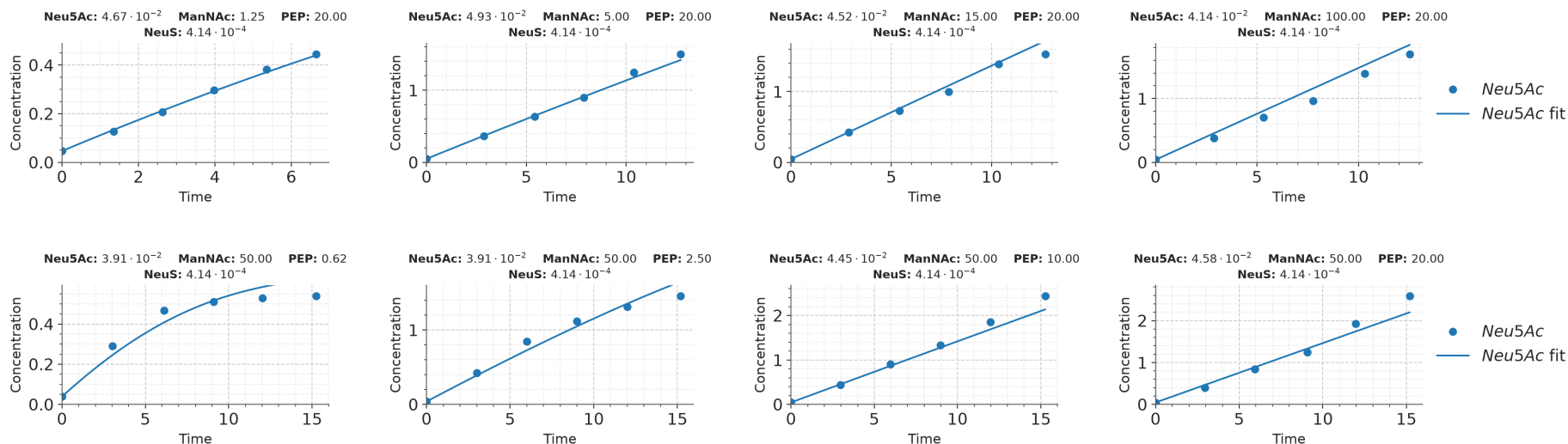
TASK 2

Task 2: kinetic parameters by Bayesian inference

$$v = k_{\text{cat}} \cdot [\text{NeuS}] \cdot [\text{ManNAc}] \cdot [\text{PEP}] / ((K_M^{\text{ManNAc}} + [\text{ManNAc}]) (K_M^{\text{PEP}} + [\text{PEP}]))$$

- **Goal:** posterior distributions of k_{cat} , K_M^{ManNAc} and K_M^{PEP} , with a fit plot
- **Model:** the enzyme is in the rate law, only Neu5Ac is measured, the metal cofactor is left out (a simplification)
- **Your decisions:** the priors, the assumed measurement noise, the sampler settings
- **Checks:** r-hat near 1, no divergences, posterior clear of the prior bounds

The reference fit, and an open question



Parameter	Posterior mean $\pm$ sd
k_{cat}	$364 \pm 6 \text{ min}^{-1}$ (6.1 s^{-1} , 9.4 U/mg)
K_M^{ManNAc}	$1.53 \pm 0.19 \text{ mM}$
K_M^{PEP}	$0.59 \pm 0.07 \text{ mM}$

- One shared k_{cat} over-predicts the ManNAc series and under-predicts the PEP series
- The two series were measured on different days and differ by roughly 25 % in activity
- What would you do next?

What to take home

- A one-page card is enough for an assistant to drive our tools
- The decisions and the checks stay with the scientist
- The cards are reusable as SKILL.md or AGENTS.md in coding agents
- What did not work for you today is the most useful feedback for us

Thank you

github.com/haeussma/enzymeml-workshop-modelling-2026

MAX HÄUBLER · UNIVERSITY OF FREIBURG

universität freiburg

